

Introduction

Football is a game of strategy. Sometimes, all it takes is one play to result in a score or change the tide of the game.

With this focal point on strategy, we developed an **MLP** to predict specific **play calls** given certain game states and a **Bayesian Linear Regression** model to predict the **expected points added (EPA)** given a play call at a certain field position.

Background & Data

Analytics are becoming increasingly leveraged by NFL to drive better decisions and outcomes. Pre-snap situational data can help intuit what the next play call is going to be and how many points you can “expect” to score.

Existing literature shows the use of MLPs to predict play calls and EPA-related statistics, but **none are as granular** in the play call predictions and **none incorporate a Bayesian-based linear regression** approach.

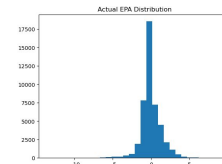
Our data contains NFL **play-by-play data** of offensive plays from the **2019 NFL season**. It contains a plethora of pre-snap information and information related to play outcomes.

Methodology

Our MLP takes in pre-snap information including **field position**, **time of game**, whether or not an offense runs a **huddle** play/is in **shotgun formation** to predict 1 of 6 defined plays. The architecture has 1 hidden layer activated by the Leaky ReLU function and output passed through the softmax function for multi-class classification.

- Left Run
- Short Left Pass
- Deep Left Pass
- Right Run
- Short Right Pass
- Deep Right Pass

The Bayesian Linear Regression approach honed in on using **field position**, **time remaining** (per quarter), **down**, and **play call** features to predict the Expected Points Added per play. We implemented a **Normal model** with Normal weight priors & Normal-Inverse Gamma for the covariance. (Based on how **actual EPA** resembles a **Gaussian**)



Results: MLP

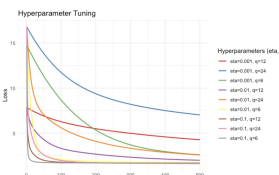
After hyperparameter tuning, the best performing MLP model yielded an accuracy of about 30.2% on test data. While **seemingly low** at face value, this **noticeably outperforms** both random choice (16.7%) and the naive baseline of always predicting the most common play type (short right pass at 25%). The model's primary confusion was predicting a short right pass when the actual play call was a short left pass.

Confusion Matrix

Predicted Play \ Actual Play	short left pass	short right pass	deep left pass	deep right pass	left run	right run
short left pass	409	417	92	118	197	155
short right pass	596	661	125	152	298	307
deep left pass	2	6	0	0	1	3
deep right pass	3	4	0	0	2	2
left run	137	156	43	33	278	289
right run	233	238	61	56	447	453

Count

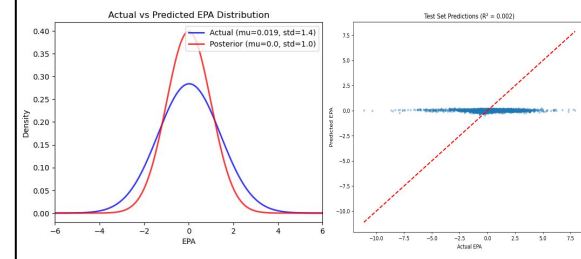
However, this is not indicative of poor model performance. Direction is usually based on defensive alignment, and since we don't have access to that in the dataset we can view it as insignificant and the **correct prediction of the crux** (short pass) is a **compliment to the model's performance**. Other missed classifications include mixing up short passes for runs and vice versa, although these are not as extreme, and indicate the model's recognition of the play's strategic intent given that both of these play types achieve similar results.



Results: Bayesian

We found that sampling was valid (convergence was reached) and that certain features moved significantly from our defined priors: **intercept**, **yard line**, and **play type**.

Our posterior distribution seems to **mirror the actual EPA** distribution, but **lacks the spread**. This looks fine, but the model's range is way off.



Conclusion & Next Steps

The Bayesian model is **not a good fit**. The R-Squared value is only partially better than **guessing the EPA per play**. To improve this, we would address **non-linearity**, **x-feature scaling**, and **categorical features** issues.

We observe that the MLP model has seemingly poor performance, but upon digging deeper, we can conclude it is able to **learn general play characteristics** and understand strategic intent behind plays from the features available. Future work would likely involve **including more features** (defensive alignment, etc), or (with the help of greater compute power), **producing a deeper neural network** that might be able to capture more nuanced trends.